

Databases for Big Data

Exercises 8: Spark / Photon

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Exercises

Exercise 1.

Describe the abstract and concrete views of the Resilient Distributed Dataset. What is a lineage in the frame of an RDD?

Exercise 2.

Present the main concepts of the Spark interface. How are the queries constructed?

Exercise 3.

Describe the concept of a Data Frame. What are the logical and physical views of DFs? How are DFs related to RDDs?

Exercise 4.

Explain the architecture of Spark SQL. What are the main building blocks? What is the role of Data Frames in Spark SQL?

Exercise 5.

Present the execution of a Data Frame. How is the logical plan of a DF converted into the physical plan?

Exercise 6.

What is Photon? Describe the architecture of Photon. How is Photon connected with the Spark query engine based on Java?